# Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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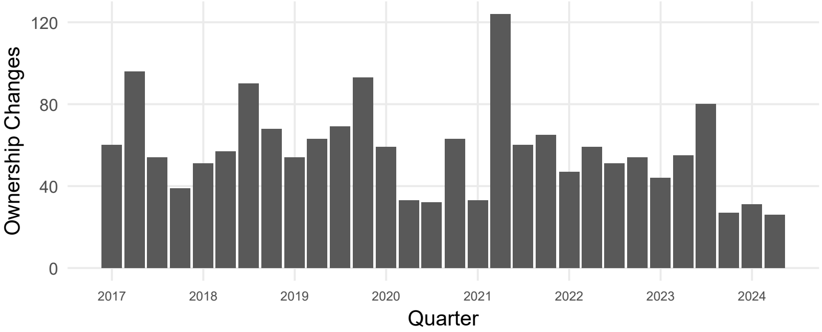
- This second type of change does **not** alter the set of providers in a market.
- Instead, effects operate through **internal production decisions**.
- This type of change is understudied in the literature, and we examine it in **nursing homes**.

# Ownership Changes in Nursing Homes by Quarter



# Why Nursing Homes?

- 15,000 U.S. nursing homes serve over 4 million people annualy
- 56% of retirement age adults will use a nursing home
- Labor is the main input into care
  - More staffing improves measured quality (Lin, 2014) and decreases deficiency citations (Harrington et al., 2000)
  - Major cost
- Staffing can easily change so it may adjust following an ownership change

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  - Do responses differ across staff type?
  - Do responses differ if a nursing home is part of a chain or not?
  - Do responses differ pre vs post-pandemic?

# Nursing Staff Types

- **Nursing homes staff care using three main nursing roles:**
  - **Registered Nurses (RNs):** clinical assessment, care planning, meds, supervision
  - **Licensed Practical Nurses (LPNs):** routine clinical tasks, meds under RN oversight
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  - **Total:** RN + LPN + CNA
- **Staffing measure: Hours Per Resident Day (HPRD)**
  - How much total time nurses spend caring for one resident over a 24-hour period
- Ownership change can affect both **levels** and **mix** which sets up cost-quality trade off.

# Cost–Quality Trade Off

- Owners can raise quality through two channels:
  - **Quantity:** increasing total nursing hours (HPRD)
  - **Skill mix:** shifting hours toward more clinically trained staff
- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed**, so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. (Park and Werner, 2011; Bowblis and Brunt, 2025)

# Why Staffing Could Rise or Fall

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressures**
  - Debt & tighter targets $\Rightarrow$ cut labor costs
- **Scenario 2: Quality focus**
  - Invest in staffing $\Rightarrow$ increase staffing
- **Scenario 3: Reorganization**
  - Shift RN/LPN/CNA composition; total hours may be similar

# Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
  - Annual cost report information
  - Contains information about costs, revenue, financial data and dates of ownership change
- **Nursing Home Compare (NHC) Archive**
  - Contains ownership information as well as facility characteristics
- **Payroll Based Journal (PBJ)**
  - Payroll-reported nurse staffing hours by staff type
  - Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, Jan 2017 – June 2024**

# Identifying Ownership Changes

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- **Challenge:** ownership is often **layered**, so an administrative "change" may not reflect a true **transfer of economic control**.
- We identify ownership changes using a **3-step process**.
- **Step 1:** HCRIS includes a variable indicating **whether and when** an ownership change occurred.
  - Useful for identifying **candidate events**, but the flag may be incomplete or capture legal restructuring rather than an economic change.
- Therefore, we have to use **NHC ownership shares** to verify and date whether control actually changes.

# Ownership Level Challenge

- **NHC**
- We identify ownership changes at the **highest level of ownership present**

| Ownership Level | Type | Owner Name | Share |
|---|---|---|---|
| *Direct ownership* | | | |
| Organization | LLC | BREDLEGS HOLDINGS LLC | 25% |
| Organization | LLC | ROCK HILL ACQUISITION III LLC | 75% |
| *Indirect ownership* | | | |
| Individual | Person | LALLY, THOMAS | 25% |
| Individual | Person | LITT, ALAN | 25% |
| Individual | Person | LITT, JONATHAN | 25% |
| Individual | Person | STURM, JOSHUA | 25% |

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- **Step 2: Validate & date using NHC**
  - Track shares
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  - Do not flag name changes or family transfers

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- **Example from the data:**
  - *Before:*
    - ESTES JAMES (84%)
    - BURCHFIELD JOHN (10%)
    - LEE CLAUDE (6%)
  - *After:*
    - JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
    - JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
  - We do **not** treat this as a meaningful ownership change

# Ownership Change Agreement

- **Step 3: Validate between HCRIS and NHC.**
- We keep facilities only if:
  - 0 changes in **both** NHC and HCRIS, or
  - Exactly 1 change in **each** dataset, within **6 months**.
- **Event month:** NHC date of ownership change.

# Cross-tab of NHC & HCRIS

| Count of changes in NHC | 0 | 1 | 2+ |
|---|---|---|---|
| 2+ | 1,137 (8.5%) | 1,377 (10.3%) | 535 (4.0%) |
| 1 | 1,874 (14.0%) | 1,589 (11.9%) | 101 (0.8%) |
| 0 | 6,217 (46.6%) | 499 (3.7%) | 15 (0.1%) |

Count of changes in HCRIS

# Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
  - RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

$$\text{HPRD}_{s,i,t} = \frac{\text{TotalHours}_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

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- **CMS implausible staffing exclusions:**
  - RN HPRD $= 0$ **and** LPN HPRD $= 0$
  - Total HPRD $< 1.5$ or $> 12$
  - CNA HPRD $> 5.25$
- **Additional restrictions:** beds $< 15$; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

# Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions (Goodman-Bacon, 2021)
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

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- **Design:** staggered-adoption DiD around ownership transitions (Goodman-Bacon, 2021)
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \, \mathbf{1}\{event\_time_{it} = \tau\} \cdot Treat_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

  - $Y_{it}$: staffing outcome in **HPRD**
  - $\beta_\tau$: effect on $Y_{it}$ at event time $\tau$
  - $X_{it}$ controls (beds, occupancy, payer mix, ownership type, chain, acuity)
  - $\alpha_i$ facility FE; $\lambda_t$ month FE
  - SEs clustered at the facility and month level

# Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

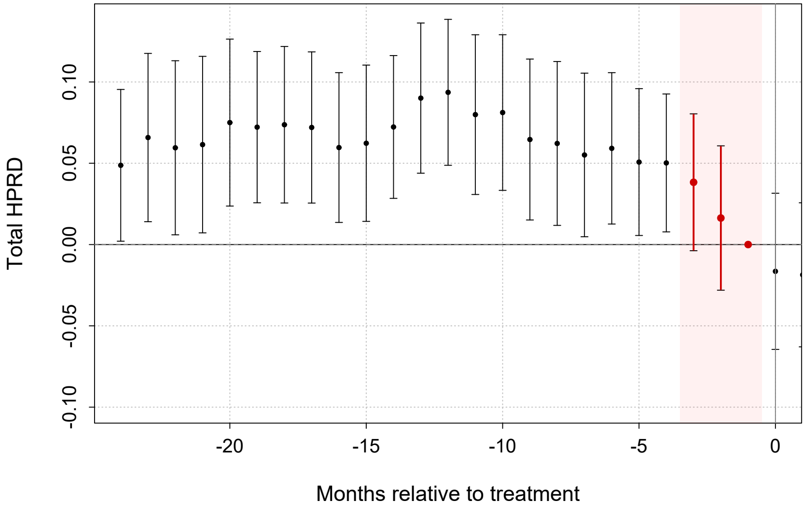
$$Y_{it} = \beta\, Post_{it} + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

  - $Post_{it} = 1$ in all months after a change
  - $\beta$: **average post-change effect** on staffing $Y_{it}$
  - Same fixed effects and controls as the event-study specification
  - SEs clustered at the facility and month level

# Identifying Assumptions

- **Parallel trends**
- **Staffing may change in anticipation of a formal ownership change**
  - A contract is signed, then there is a gap until the closing/transfer date.
  - During this gap, staffing may adjust
  - We want to observe the date the contract was signed.
- **Implication:** The months in this window may violate parallel trends.

# Evidence of violation in the Pre-Period



# Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.

- **Implementation**
  - Exclude $\tau \in \{-3, -2, -1\}$
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- **Implementation**
  - Exclude $\tau \in \{-3, -2, -1\}$
  - Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0: \ \beta_\tau = 0 \quad \forall \, \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

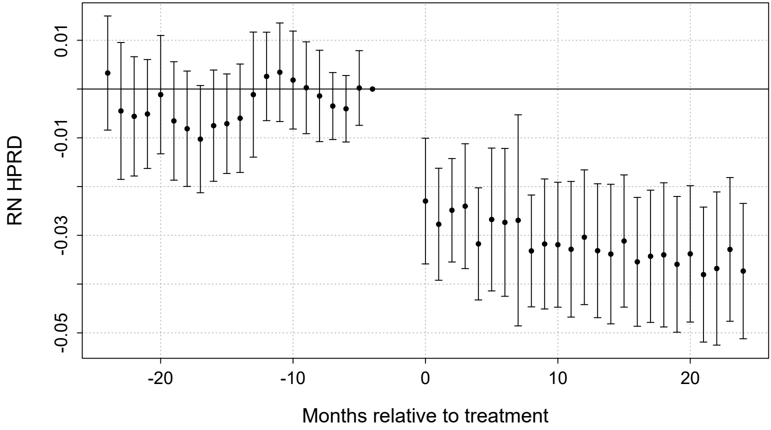
# Parallel Trends Tests (Joint Wald) — Total Staffing

Table: **Pre-trend Wald Test p-values (Total Staffing)**

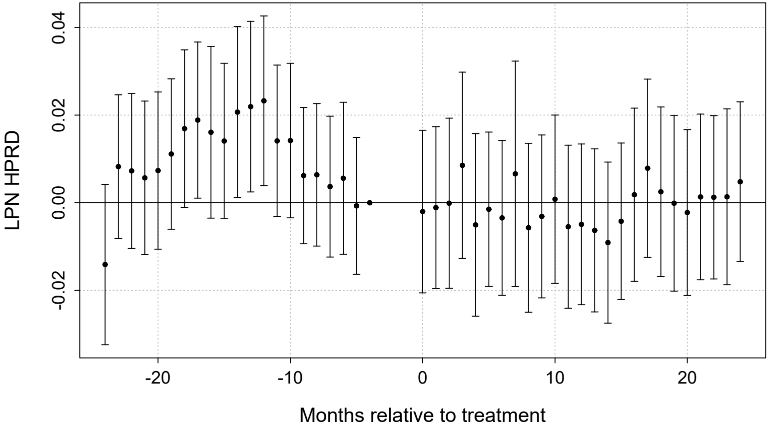
| | **Total (Levels)** | **Total (Logs)** |
|---|---|---|
| 2 Year Full Pre-Window | 0.0006 | 0.0012 |
| 2 Year Window with Donut | 0.2046 | 0.3268 |
| 1 Year Window with Donut | 0.6387 | 0.6102 |

*Notes:* Joint Wald $\chi^2$ test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.
Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

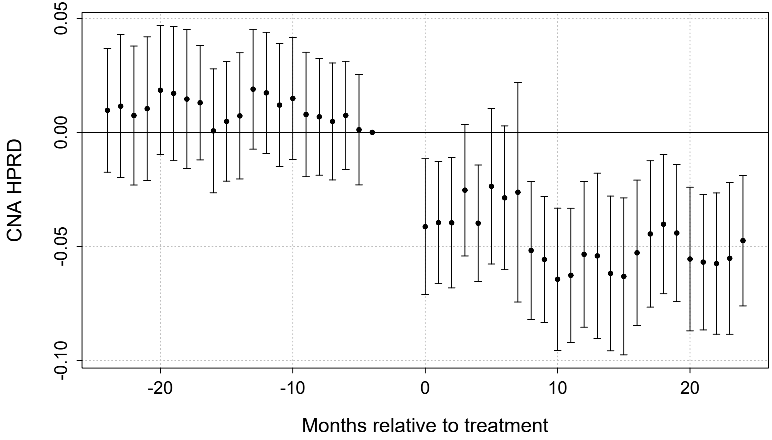
# Event Study Results (RN HPRD)



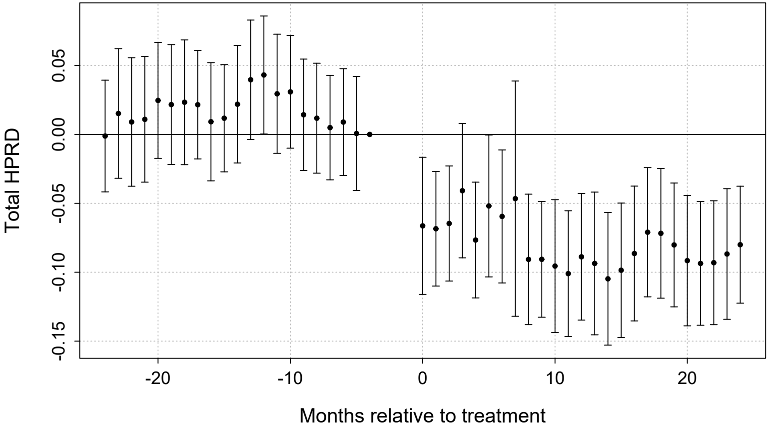
# Event Study Results (LPN HPRD)



# Event Study Results (CNA HPRD)



# Event Study Results (Total Nursing HPRD)



# TWFE DiD — Baseline (RN)

| | RN | LPN | CNA | Total |
|---|---|---|---|---|
| **HPRD** | $-0.032^{***}$ | 0.000 | $-0.054^{***}$ | $-0.086^{***}$ |
| | (0.005) | (0.006) | (0.009) | (0.013) |
| **Log(HPRD)** | $-0.092^{***}$ | 0.003 | $-0.028^{***}$ | $-0.026^{***}$ |
| | (0.016) | (0.009) | (0.005) | (0.004) |

*Notes:* Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628{,}107$

# TWFE DiD — Baseline (LPN)

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| **HPRD** | $-0.032^{***}$ | 0.000 | $-0.054^{***}$ | $-0.086^{***}$ |
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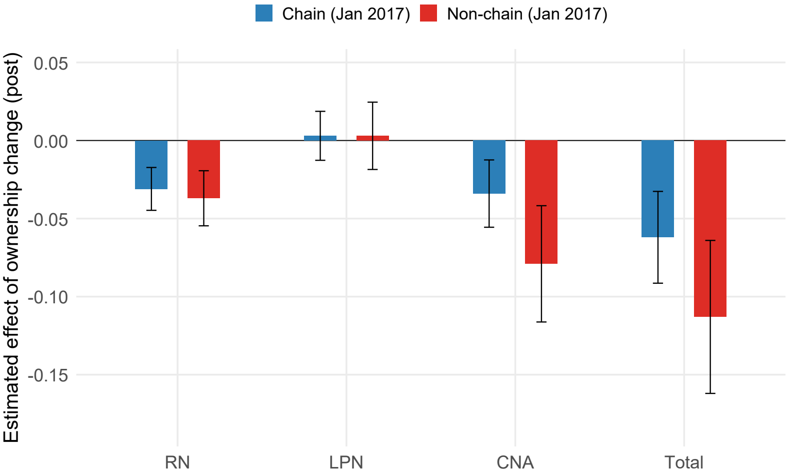
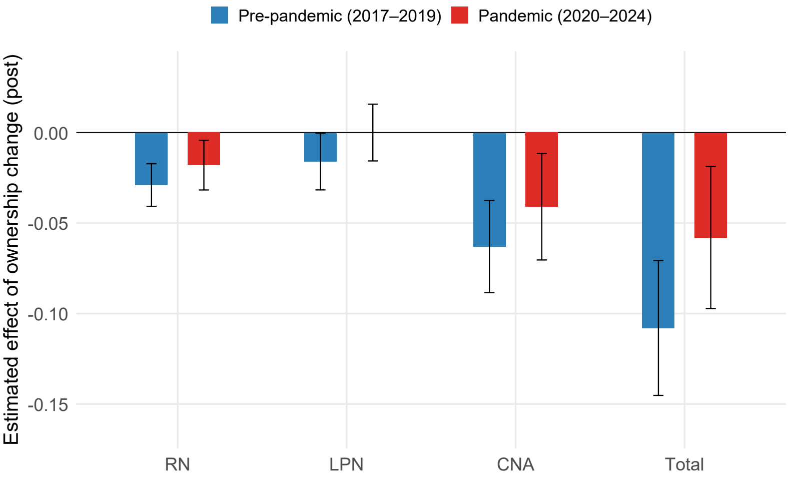
# Heterogeneity: Chain vs Non-Chain Facilities




Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

# Heterogeneity: Pre vs Post-Pandemic




Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

# Robustness Checks

- **Stacked Difference-in-Difference**
- **Smaller and wider Donut Hole**
- **Gaps in our Data.**

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- **Effects are heterogeneous across settings.**
  - Larger declines **pre-pandemic** vs. **pandemic period**
  - Larger declines for **non-chain** vs. **chain** facilities

# Discussion

- **Key finding:** RN, CNA, and Total hours fall after ownership change; **LPN hours do not.**

- **Why might LPN hours not decline?**
  - Compliance
  - Cost
  - Substitution

*(e.g., Lin 2014; Harrington et al. 2000; Bowblis & Ghattas 2017)*

# Discussion

- **Question:** Are our findings economically **meaningful**?

| Staff type | Sample Average | Effects |
|---|---|---|
| RN | 25.6 min/day | −1.9 min/day |
| LPN | 47.8 min/day | 0.0 min/day |
| CNA | 2 hr 6 min/day | −3.2 min/day |
| Total | 3 hr 19 min/day | −5.2 min/day |

- Lin (2014) finds 0.3 HPRD ($\approx$ 18 minutes) increase in RN staffing $\rightarrow$ improves deficiencies 16% .
  - Our results would translate to a 1.6% increase in deficiencies.

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- **Why are declines larger in non-chain facilities?**
  - Small business
  - Local
- **Why are declines smaller after Covid?**
  - Labor market shocks
  - Higher oversight/compliance
  - Tighter constraints

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  - Linking our findings to quality metrics and resident outcomes
  - Identifying mechanisms

# Thank you

Questions?

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